

Demo for Data and Network Security report

Roberto Di Rosa, 2043388, Sapienza University of Rome,
Name Surname, 1000000, Sapienza University of Rome,

Abstract—As mobile devices become ubiquitous, the resulting cellular telemetry forms an indelible human footprint. Modern forensic frameworks such as SURE and MoCo exploit behavioral metadata to bridge the gap between anonymity and identity, rendering traditional anonymization techniques like k-anonymity ineffective. This report analyzes these tracking frameworks and proposes the adoption of a parallel radio network based on the Meshtastic protocol as the only robust mitigation. By bypassing centralized cellular operator data systems, we demonstrate that operational security can be maintained even against deep-learning-driven surveillance.

Index Terms—Machine Learning, Deep Learning, Meshtastic, User Re-identification, Operational Security, LoRa.



1 Introduction

In the modern digital landscape, data has become a literal battlefield where every wireless transmission leaves a trace [1]. As long as a mobile connection is maintained, a device generates a continuous stream of telemetry that forms an indelible human footprint [1]. This footprint is captured passively by Cellular Network Operator Data Systems, which aggregate vast quantities of metadata regarding service requests, data exchanges, and handovers between tracking areas [2, 3].

1.1 The Problem—Behavior as Identity

The fundamental challenge is that human habits do not change when a name is erased [1]. Modern forensic frameworks like SURE (Smartphone Users RE-identification) and MoCo (Mobile Contact Detection) treat behavior as identity [2, 4]. By extracting “pure behavioral signatures” from sequences of Base Station (BS) associations and traffic levels, these frameworks can uniquely identify a user [1, 3]. Research indicates that over 60% of user traffic is concentrated on just two antennas—the “Top 2 BS” rule—which serves as a spatial anchor for identifying a target’s home or workplace [1, 3].

1.2 The Illusion of Anonymity

Traditional security methods, such as k-anonymity and the suppression of identifiers like IMSI, are “blunt weapons” against deep learning [1, 3]. These methods fail because they only obscure identifying fields while leaving underlying spatio-temporal metadata unchanged [1, 3]. Dual-encoder architectures can mathematically bridge the gap between an anonymous index and a known identity with AUC scores above 0.9 using only 4 days of data [2, 3].

1.3 Project Objectives

This report demonstrates that as long as a user remains on a cellular grid, full operational disclosure is a mathematical certainty [1, 3]. We propose bypassing the cellular operator data system entirely by deploying a parallel radio network built on the Meshtastic protocol [5, 6].

2 Background

2.1 Deep Learning Networks

Complex computational problems can be addressed through various methodologies, prominent among which is Deep Learning. This paradigm is fundamentally rooted in mathematical optimization. Consider a predictive model defined as:

$$\begin{aligned} f_{\theta} : \mathbb{R}^K &\rightarrow \mathbb{R}^m \\ \forall x \in \mathbb{R}^K, \Theta(x) &\in \mathbb{R}^m \end{aligned} \quad (1)$$

The objective of training is to optimize the internal parameters—specifically the weights and biases of the neurons—to maximize the model’s predictive accuracy.

This optimization process is systematically executed through the following iterative phases:

- **Initialization:** The network parameters are initially set using randomized values or predefined distribution strategies.
- **Forward Propagation:** An input vector x is processed through the mapping function to generate a predicted output, denoted as $\hat{y}$ (commonly referred to as the forward pass).
- **Error Evaluation:** The discrepancy between the predicted output and the ground-truth label y is quantified by evaluating a Loss function $\mathbf{L}$ (defined in Equation 2).
- **Backpropagation and Optimization:** Utilizing the backpropagation algorithm, the gradient of the loss function is computed with respect to each network weight via the chain rule. The model parameters are subsequently adjusted in the negative gradient direction—a process known as gradient descent—effectively minimizing the objective function.

$$\begin{aligned} \hat{y} &= \Theta(x) \\ L(x) &= y - \hat{y} \end{aligned} \quad (2)$$

While the mathematical formulation of the loss function varies depending on the specific task architecture, its underlying objective remains constant: the resulting scalar serves

as a performance metric utilized by optimization algorithms to converge toward the global minimum¹.

The foundational principles detailed above describe standard Machine Learning. The distinction between classical Machine Learning and Deep Learning primarily lies in structural complexity; standard Machine Learning models typically rely on shallow architectures, whereas Deep Learning utilizes deep neural networks characterized by multiple successive hidden layers.

Formally, Deep Learning $\subset$ Machine Learning. While both paradigms maintain distinct domains of utility, this report focuses primarily on the deployment and behavior of Deep Learning architectures.

2.2 Base Station

This infrastructure comprises cellular base stations maintained by telecommunication service providers. These stations aggregate vast quantities of telemetry and location data which, despite the application of anonymization techniques, remain vulnerable to re-identification attacks orchestrated via deep learning models. Consequently, any adversary possessing sufficient computational resources could exploit these vulnerabilities to execute large-scale mass surveillance².

2.3 The SURE Framework (User Re-identification)

The SURE (Smartphone Users RE-identification) framework is a deep learning model designed to accurately classify whether two input network traces belong to the same user. It utilizes a parallel dual-encoder architecture to process anonymous and known-identity traces independently, preventing data contamination before mathematical comparison. The core of the system is spatio-temporal embedding, which vectorizes four parameters: Base Station (BS) IDs, association duration, and discretized levels of uplink and downlink traffic. Evaluations on datasets involving 10,000 users show that SURE achieves a surgical AUC above 0.9 even with a data window as narrow as four days.

2.4 The MoCo Framework (Contact Detection)

MoCo (Mobile Contact detection) focuses on identifying users moving in "lockstep," such as those co-traveling in the same vehicle. The framework uses a three-stage pipeline: data denoising via the DBSCAN algorithm to remove remote BS noise, a spatio-temporal filter (ST-Filter) to eliminate unlikely contact pairs (typically requiring $\geq 60\%$ spatial similarity), and a learning network that integrates an LSTM-based spatial branch with an MLP-based temporal branch. MoCo can effectively map a target's "inner circle" by searching for traces that mirror their movements, reaching an average F1-score of 87.13%.

2.5 Vulnerabilities in Self-Supervised Learning (SSL)

While powerful, these frameworks are susceptible to backdoor attacks [7, 8]. An adversary can poison as little as 0.5% of unlabeled training data with a rigid "trigger" patch, causing the model to associate that trigger with a target category at test time [3, 9].

1. The parameter configuration that minimizes the loss value.
2. Amnesty International: An Easy Guide to Mass Surveillance

3 Overview of your proposed approach

3.1 Technical Architecture

We propose Meshtastic, a decentralized mesh network utilizing the LoRa physical layer and Chirp Spread Spectrum (CSS) [5]. Signals consist of frequency-sweeping "chirps" with a 16-up-chirp Preamble for synchronization and a specific Sync Word (0x2B) to distinguish the network [5, 10].

3.2 Decentralized Metadata Protection

The primary strength is the total bypass of Cellular Operator Data Systems [5]. In a mesh architecture, there is no central database logging "Top 2 BS" anchors or hourly traffic windows—the features required by SURE and MoCo for re-identification [3, 5].

3.3 Encryption and Routing

Meshtastic payloads are encrypted using AES-128 shared-key encryption [5, 11]. Only nodes with the correct channel key can interpret data, while decentralized "hop limits" prevent the creation of a trackable historical footprint [5, 12].

TABLE 1
SWOT Analysis: Meshtastic Parallel Network vs. Cellular

	Internal	External
Strengths	No centralized operator logs; metadata privacy; AES encryption [3, 5].	Opportunities: High-privacy operational demand; resistance to state-level AI [6].
Weaknesses	Lower data rates; limited propagation range of LoRa [5, 13].	Threats: Signal triangulation; future RF fingerprinting evolutions [6].

4 Evaluation

4.1 Proposed Experimental Setup

We propose comparing a control group on a cellular network against a test group using Meshtastic over 7 days [3, 6]. We will apply the SURE architecture to match anonymous traces back to a leaked identity database [2, 3].

4.2 Performance Metrics

Success will be measured via AUC, F1-Score, and Precision-Recall [2, 4]. A successful mitigation will show the Meshtastic group's re-identification AUC dropping to near-random levels compared to the control group's ≥ 0.9 [2, 3].

5 Related Work

5.1 The Collapse of K-Anonymity

Removing names is useless if spatial behavior—the "Top 2 BS" routine—remains stable [1, 3]. Mathematical proof shows male targets and those over 45 are the easiest to track due to stable logistical routines [1, 3].

5.2 Alternative SSL Defenses

Techniques like ASTRA (Adaptive-Attacks) and RoCL (Robust Contrastive Learning) harden SSL models against perturbations [14, 15]. However, these do not address the root cause of tracking: centralized metadata logging [3, 6].

6 Related work

In this section, you need to analyze your direct competitors (e.g. [1] if you talk about watermarking), explaining a bit what they do and why you think your approach is better. What has been tried before?

7 Conclusions

As long as cellular connections are maintained, full operational disclosure is inevitable due to behavioral predictability [1, 3]. Parallel radio networks like Meshtastic are not just alternatives but technical necessities for operating outside the "Battlefield of Data" [1, 5].

References

- [1] Y. Adi, C. Baum, M. Cisse, B. Pinkas, and J. Keshet, "Turning your weakness into a strength: Watermarking deep neural networks by backdooring," in 27th USENIX Security Symposium (USENIX Security 18). Baltimore, MD: USENIX Association, Aug. 2018, pp. 1615–1631.